Element B: Documentation and Analysis of Prior Design Attempts

What is being done now with the problem? What is being done in other places with a similar situation: Ms. Tatro (the primary stakeholder) needs more bookends to store books in the ERHS library because the current supply is insufficient, and purchasing more is expensive. By designing and construction bookends from recycled materials, we can provide a cost-effective and sustainable solution. Currently, the ERHS library has some bookends, but they are not enough to keep books properly organized, purchasing additional bookends is costly, making it necessary to explore alternative solutions. Other schools and libraries facing similar issues have implemented creative methods such as repurposing materials, using DIY, bookends made from recycled items ore designing cost-effective 3D printed alternatives. These approaches reduce costs while ensuring books remain organized and accessible.

Documentation of Prior Attempts

1. **3d-Printed Bookends**
   Description: Libraries and Individuals with access to 3D printed technology have designed and printed custom bookends. This method allows for the creation of unique and tailored designs. Prinatbles.com week's 3D printing competition is all about creating the ultimate bookends for one's personal library.
   Successes: 3D printing offers precision and the ability to produce bookends that fit specific deimesnsions and aesthetic preferences competetions and online platforms provide numerous designs for inspiration.
   Failures: The initial investment in a 3D printer can be costly. Additionally, the printing process requires technical knowledge and time, which may not be feasible for all users. Also, since the bookends were for one's personal library, it may not be strong enough to hold all the books in a public library, rather than a small one in someone's house.
   Source: Bookends | Contest theme | Printables.com. (2023). Printables.com. https://www.printables.com/contest/378-bookends

2. **Diy Bookends from recycled materials**

- **Description:** Individuals have crafted bookends using recycled materials such as cardboard, wood, and fabric. These DIY projects often involve repurposing household items to create functional and decorative bookends.
- **Successes:** This approach is cost effective and enrivonmetally friendly. It allows for customization to match personal or library aesthetics. Tutorials and guides are readily available, making the process accessible.
- **Failures:** The durability and stability of DIY bookends can vary depending on the materials used. Lightweight materials may not support heavier books effectively, leading to potential collapse or damage. Also, since the bookends were for one's personal library,

it may not be strong enough to hold all the books in a public library, rather than a small one in someone's house.

- **Source:** Delight Your Inner Bookworm With 15 DIY Bookends. (n.d.). The Spruce Crafts. https://www.thesprucecrafts.com/diy-bookends-tutorials-4691862

3. **Using Heavy objects as bookends.**

- **Description:** Some have used heavy objects (i.e, concrete) as makeshift bookends to keep books upright.
- **Successes:** This method is straightforward and utilizies readily available items eliminating additional costs, it can be effective for supporting books on sturdy shelves.
- **Failures:** Not all heavy objects are suitable for bookends, they can lack shape and stability. They also might not provide the desired aesthetic or could pose risks if they are unstable, breakable, or too heavy.
- **Source:** (2014, December 14). What is a good makeshift bookend? Lifehacks Stack Exchange.https://lifehacks.stackexchange.com/questions/1775/what-is-a-good-makeshift-bookend

Conclusion: Analyzing these prior attempts reveals that while there are multiple creative solutions to the problem of insufficient bookends, each comes with its own set of advantages and limitations. DIY bookends from recycled materials offer a sustainable and customizable option but may lack durability. Using heavy objects is cost-effective but may not always provide the desired stability or appearance. 3D printing allows for precise and personalized designs but requires access to specific technology and expertise. Therefore, the chosen solution should balance cost, sustainability, durability, and aesthetic considerations to effectively address the needs of the ERHS library.

Reference Citations:

- Lifehacks Stack Exchange, 2013
  Printables.com 2023
- The Spruce Crafts, 2019-2020